CT PERFORMANCE EVALUATION REPORT

PRE-TEST / POST FAN CHANGE / POST DISTRIBUTION CHANGE

— THREE-STAGE SINGLE CELL ASSESSMENT —

CT - 02 , CELL 02

OWNER / CLIENT	DHARIWAL INFRASTRUCTURE LTD CHANDRAPUR
TESTING AGENCY	SS COOLING TOWER CONSULTANTS (SSCTC)
DATE OF TEST	27 March 2026
REPORT DATE	15 April 2026
ASSESSMENT METHOD	CTI ATC-105 Merkel Method — Applied independently to each of 3 test stages

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PREAMBLE

This report documents the performance evaluation of the cooling tower.

Tests were conducted as per CTI ATC-105 standard Merkel method.

MEMBERS PRESENT FOR THE TEST

CLIENT

- Mr. Sharma
- Mr. Verma

SS COOLING TOWER CONSULTANTS (SSCTC)

- Eng. Kumar
- Tech. Rajan

ASSESSMENT METHOD & ANALYSIS

CTI ATC-105 Merkel method was applied.

Three stage evaluation performed.

INSTRUMENT PLACEMENT AND READINGS

Wet bulb temperature sensors placed at inlet.

Flow meters installed.

CONCLUSION AND REMARKS

1. The cooling tower shows a shortfall in performance.
2. Post-fan-change test shows improvement.

SUGGESTIONS AND SPECIAL NOTES

1. Replace fan blades annually.
2. Clean distribution nozzles quarterly.

FINAL DATA AND CONCLUSION OF PRE AND POST TEST

The following table presents a consolidated comparison of all measured and calculated parameters across the three test stages conducted during this evaluation.

Parameter	Unit	TEST 1 Pre-Test	TEST 2 Post Fan Change	TEST 3 Post Distr.
Water Flow	m3/hr	1200	1210	1215
Hot Water Temp	degC	42.5	41.8	41.2
Cold Water Temp	degC	32.1	31.5	31.0

NOTES

1. All temperatures in degrees Celsius.
2. Flow rates in cubic metres per hour.

ATC-105 PERFORMANCE EVALUATION — TEST 1 — PRE TEST

DESIGN VS RECORDED TEST CONDITIONS

Parameter	Unit	Design Conditions	Recorded Test Conditions
Water Flow	m³/hr	1200.00	1185.00
Hot Water Temp	°C	43.00	42.50
Cold Water Temp	°C	31.00	32.10
Wet Bulb Temp	°C	27.00	26.80
Range	°C	12.00	11.80
Approach	°C	4.00	5.30

STEP 1

TABLE 1 — CWT vs Range at Test WBT

Using the Merkel method, predicted Cold Water Temperatures (CWT) are computed at three fan speeds (90%, 100%, 110% of design flow) for three range values (80%, 100%, 120% of design range) at the recorded Test Wet Bulb Temperature.

Range %	Range (°C)	CWT @ 90% Flow	CWT @ 100% Flow	CWT @ 110% Flow
80%	9.44 °C	31.20	30.50	29.90
100%	11.8 °C	32.10	31.40	30.80
120%	14.16 °C	33.00	32.30	31.70

STEP 2 — CROSS PLOT 1 — CWT vs Range at Test WBT

Plot the three CWT lines (90%, 100%, 110% flow) from Table 1 against Range. Draw a vertical line at the Test Range to read off the CWT for each flow condition. These three (Flow, CWT) pairs form Table 2.

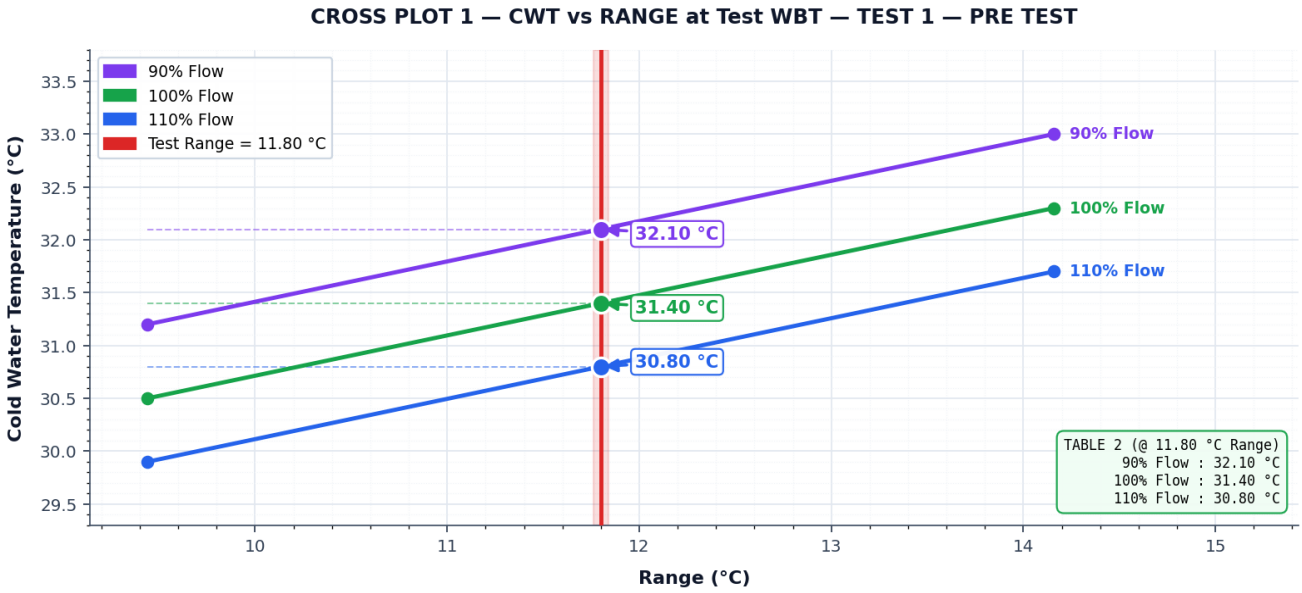

Figure 1: Cross Plot 1 — CWT vs Range at Test WBT — TEST 1 — PRE TEST

TABLE 2 — CWT at Test Range = 11.80 °C

Flow Condition	Water Flow (m³/hr)	CWT at Range 11.80°C
90% Flow	1080.00	32.10
100% Flow	1200.00	31.40
110% Flow	1320.00	30.80

STEP 3 — ADJUSTED WATER FLOW — Air Density Correction

Adjust the recorded water flow for air density differences between test and design conditions: Q_adj = Q_test × (Air Density Ratio)^0.5. The density ratio accounts for atmospheric variation in air mass flow delivered by the fan.

i. Recorded Test Water Flow (Q_test)	**1185.00 m³/hr**
ii. Air Density Ratio (ρ_design / ρ_test)	**1.00**
iii. CWT @ 90% Design Flow (Cross Plot 1)	**32.10 °C**
iv. CWT @ 100% Design Flow (Cross Plot 1)	**31.40 °C**
v. CWT @ 110% Design Flow (Cross Plot 1)	**30.80 °C**
vi. Test Hot Water Temperature (HWT)	**42.50 °C**
vii. Test Cold Water Temperature (CWT)	**32.10 °C**
viii. Test Wet Bulb Temperature (WBT)	**26.80 °C**
ix. Fan Power — Design	**45.00 kW**
x. Fan Power — Test	**44.20 kW**
Q_adj — Adjusted Water Flow (Density Corrected)	**1192.00 m³/hr**

STEP 4 — CROSS PLOT 2 — Water Flow vs CWT (Design WBT & Range)

Plot (Water Flow, CWT) pairs from Table 2. Locate Q_adj on the performance curve to read the Predicted CWT. Then find the water flow at which the curve intersects the Design CWT — this is the Predicted Flow used in Step 5.

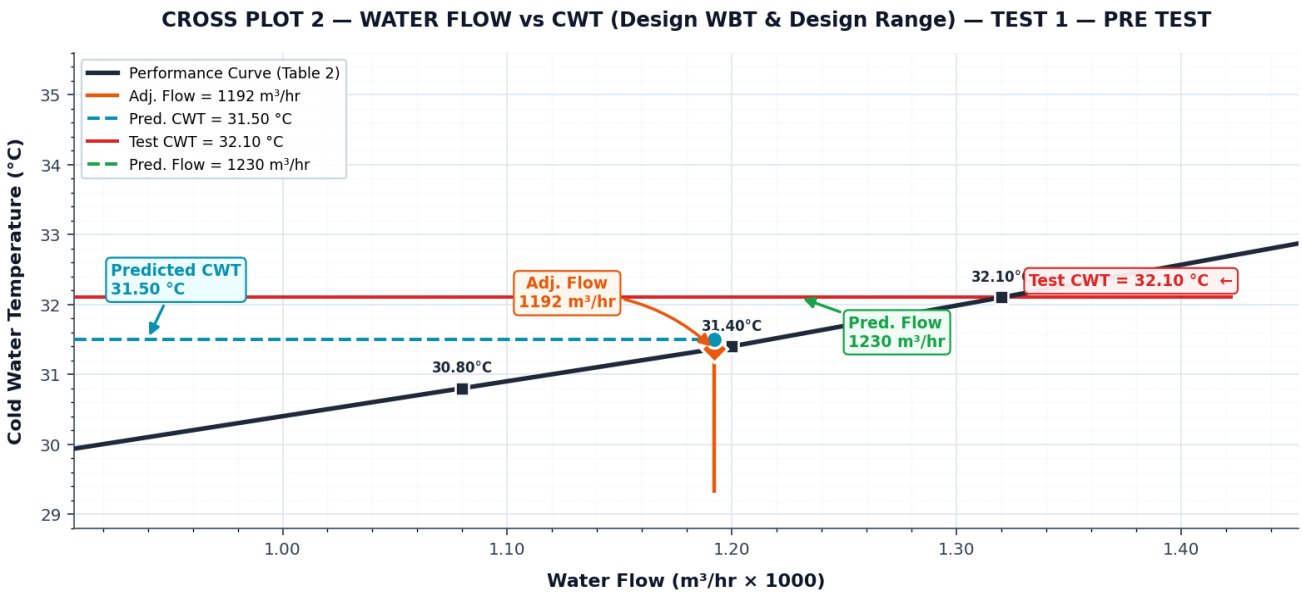

Figure 2: Cross Plot 2 — Water Flow vs CWT (Design WBT & Range) — TEST 1 — PRE TEST

FINAL ATC-105 RESULTS — TEST 1 — PRE TEST

STEP 5

FINAL EVALUATION — CWT Shortfall & Capability

Compare the Predicted CWT (from Cross Plot 2 at Q_adj) with the Test CWT. CWT Shortfall = Test CWT – Predicted CWT. A negative shortfall means the tower performed below specification. Capability % = (Q_adj / Predicted Flow) × 100.

Step	Parameter	Value
i	Q_adj — Adjusted Water Flow (density corrected)	1192.00 m³/hr
ii	Predicted CWT — from Cross Plot 2 at Q_adj	31.50 °C
iii	Test CWT — Recorded during test	32.10 °C
iv	Predicted Water Flow — at Design CWT on Cross Plot 2	1230.00 m³/hr
v	CWT Shortfall = Test CWT – Predicted CWT	0.60 °C
vi	Cooling Tower Capability = (Q_adj / Pred. Flow) × 100	96.90 %
vii	Air Density Ratio applied	1.00

CONCLUSION — TEST 1 — PRE TEST

CWT SHORTFALL IN PERFORMANCE	COOLING TOWER CAPABILITY
+0.60 °C	**96.9 %**
Test CWT (32.10°C) vs Predicted CWT (31.50°C)	= 1192 / 1230 × 100

Tower EXCEEDS design by 0.60°C — Capability: 96.9%

ATC-105 PERFORMANCE EVALUATION — TEST 2 — POST FAN CHANGE

DESIGN VS RECORDED TEST CONDITIONS

Parameter	Unit	Design Conditions	Recorded Test Conditions
Water Flow	m³/hr	1200.00	1185.00
Hot Water Temp	°C	43.00	42.50
Cold Water Temp	°C	31.00	31.80
Wet Bulb Temp	°C	27.00	26.80
Range	°C	12.00	11.80
Approach	°C	4.00	5.00

STEP 1

TABLE 1 — CWT vs Range at Test WBT

Using the Merkel method, predicted Cold Water Temperatures (CWT) are computed at three fan speeds (90%, 100%, 110% of design flow) for three range values (80%, 100%, 120% of design range) at the recorded Test Wet Bulb Temperature.

Range %	Range (°C)	CWT @ 90% Flow	CWT @ 100% Flow	CWT @ 110% Flow
80%	9.44 °C	31.20	30.50	29.90
100%	11.8 °C	32.10	31.40	30.80
120%	14.16 °C	33.00	32.30	31.70

STEP 2

CROSS PLOT 1 — CWT vs Range at Test WBT

Plot the three CWT lines (90%, 100%, 110% flow) from Table 1 against Range. Draw a vertical line at the Test Range to read off the CWT for each flow condition. These three (Flow, CWT) pairs form Table 2.

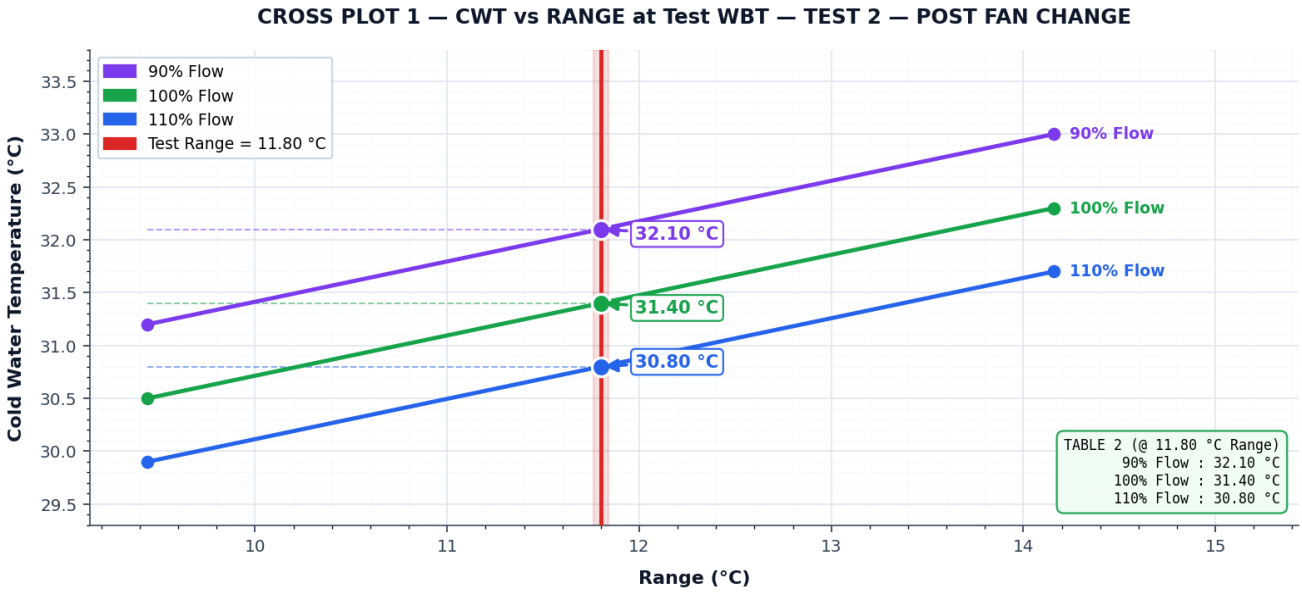

Figure 1: Cross Plot 1 — CWT vs Range at Test WBT — TEST 2 — POST FAN CHANGE

TABLE 2 — CWT at Test Range = 11.80 °C

Flow Condition	Water Flow (m³/hr)	CWT at Range 11.80°C
90% Flow	1080.00	32.10
100% Flow	1200.00	31.40
110% Flow	1320.00	30.80

STEP 3 — ADJUSTED WATER FLOW — Air Density Correction

Adjust the recorded water flow for air density differences between test and design conditions: Q_adj = Q_test × (Air Density Ratio)^0.5. The density ratio accounts for atmospheric variation in air mass flow delivered by the fan.

i. Recorded Test Water Flow (Q_test)	**1185.00 m³/hr**
ii. Air Density Ratio (ρ_design / ρ_test)	**1.00**
iii. CWT @ 90% Design Flow (Cross Plot 1)	**32.10 °C**
iv. CWT @ 100% Design Flow (Cross Plot 1)	**31.40 °C**
v. CWT @ 110% Design Flow (Cross Plot 1)	**30.80 °C**
vi. Test Hot Water Temperature (HWT)	**42.50 °C**
vii. Test Cold Water Temperature (CWT)	**31.80 °C**
viii. Test Wet Bulb Temperature (WBT)	**26.80 °C**
ix. Fan Power — Design	**45.00 kW**
x. Fan Power — Test	**44.20 kW**
Q_adj — Adjusted Water Flow (Density Corrected)	**1192.00 m³/hr**

STEP 4 — CROSS PLOT 2 — Water Flow vs CWT (Design WBT & Range)

Plot (Water Flow, CWT) pairs from Table 2. Locate Q_adj on the performance curve to read the Predicted CWT. Then find the water flow at which the curve intersects the Design CWT — this is the Predicted Flow used in Step 5.

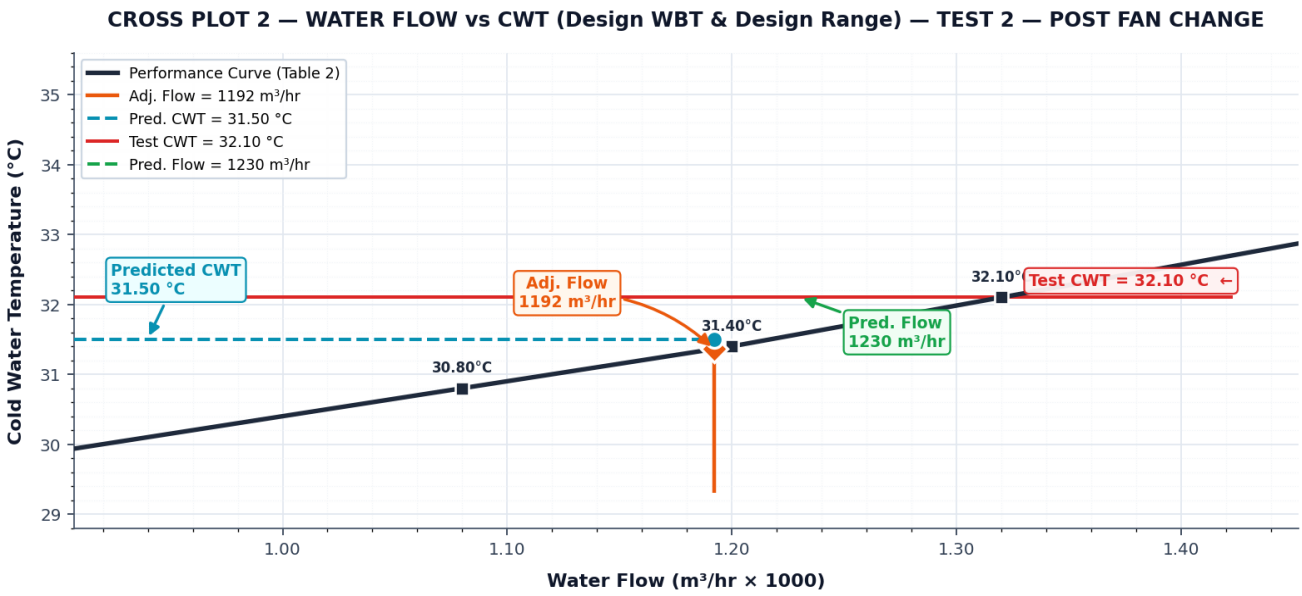

Figure 2: Cross Plot 2 — Water Flow vs CWT (Design WBT & Range) — TEST 2 — POST FAN CHANGE

FINAL ATC-105 RESULTS — TEST 2 — POST FAN CHANGE

STEP 5

FINAL EVALUATION — CWT Shortfall & Capability

Compare the Predicted CWT (from Cross Plot 2 at Q_adj) with the Test CWT. CWT Shortfall = Test CWT – Predicted CWT. A negative shortfall means the tower performed below specification. Capability % = (Q_adj / Predicted Flow) × 100.

Step	Parameter	Value
i	Q_adj — Adjusted Water Flow (density corrected)	1192.00 m³/hr
ii	Predicted CWT — from Cross Plot 2 at Q_adj	31.50 °C
iii	Test CWT — Recorded during test	31.80 °C
iv	Predicted Water Flow — at Design CWT on Cross Plot 2	1230.00 m³/hr
v	CWT Shortfall = Test CWT – Predicted CWT	0.30 °C
vi	Cooling Tower Capability = (Q_adj / Pred. Flow) × 100	96.90 %
vii	Air Density Ratio applied	1.00

CONCLUSION — TEST 2 — POST FAN CHANGE

CWT SHORTFALL IN PERFORMANCE	COOLING TOWER CAPABILITY
+0.30 °C	**96.9 %**
Test CWT (31.80°C) vs Predicted CWT (31.50°C)	= 1192 / 1230 × 100

Tower EXCEEDS design by 0.30°C — Capability: 96.9%

ATC-105 PERFORMANCE EVALUATION — TEST 3 — POST DISTRIBUTION CHANGE

DESIGN VS RECORDED TEST CONDITIONS

Parameter	Unit	Design Conditions	Recorded Test Conditions
Water Flow	m³/hr	1200.00	1185.00
Hot Water Temp	°C	43.00	42.50
Cold Water Temp	°C	31.00	31.30
Wet Bulb Temp	°C	27.00	26.80
Range	°C	12.00	11.80
Approach	°C	4.00	4.50

STEP 1

TABLE 1 — CWT vs Range at Test WBT

Using the Merkel method, predicted Cold Water Temperatures (CWT) are computed at three fan speeds (90%, 100%, 110% of design flow) for three range values (80%, 100%, 120% of design range) at the recorded Test Wet Bulb Temperature.

Range %	Range (°C)	CWT @ 90% Flow	CWT @ 100% Flow	CWT @ 110% Flow
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STEP 2

CROSS PLOT 1 — CWT vs Range at Test WBT

Plot the three CWT lines (90%, 100%, 110% flow) from Table 1 against Range. Draw a vertical line at the Test Range to read off the CWT for each flow condition. These three (Flow, CWT) pairs form Table 2.

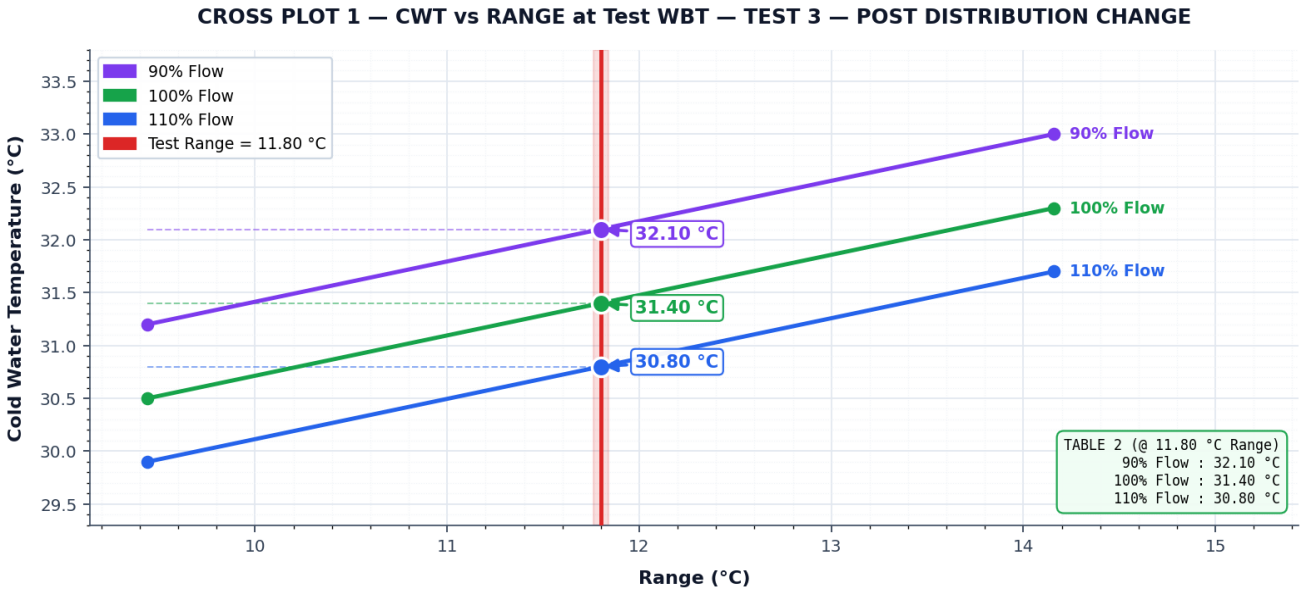

Figure 1: Cross Plot 1 — CWT vs Range at Test WBT — TEST 3 — POST DISTRIBUTION CHANGE

TABLE 2 — CWT at Test Range = 11.80 °C

Flow Condition	Water Flow (m³/hr)	CWT at Range 11.80°C
90% Flow	1080.00	32.10
100% Flow	1200.00	31.40
110% Flow	1320.00	30.80

STEP 3 — ADJUSTED WATER FLOW — Air Density Correction

Adjust the recorded water flow for air density differences between test and design conditions: Q_adj = Q_test × (Air Density Ratio)^0.5. The density ratio accounts for atmospheric variation in air mass flow delivered by the fan.

i. Recorded Test Water Flow (Q_test)	**1185.00 m³/hr**
ii. Air Density Ratio (ρ_design / ρ_test)	**1.00**
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Q_adj — Adjusted Water Flow (Density Corrected)	**1192.00 m³/hr**

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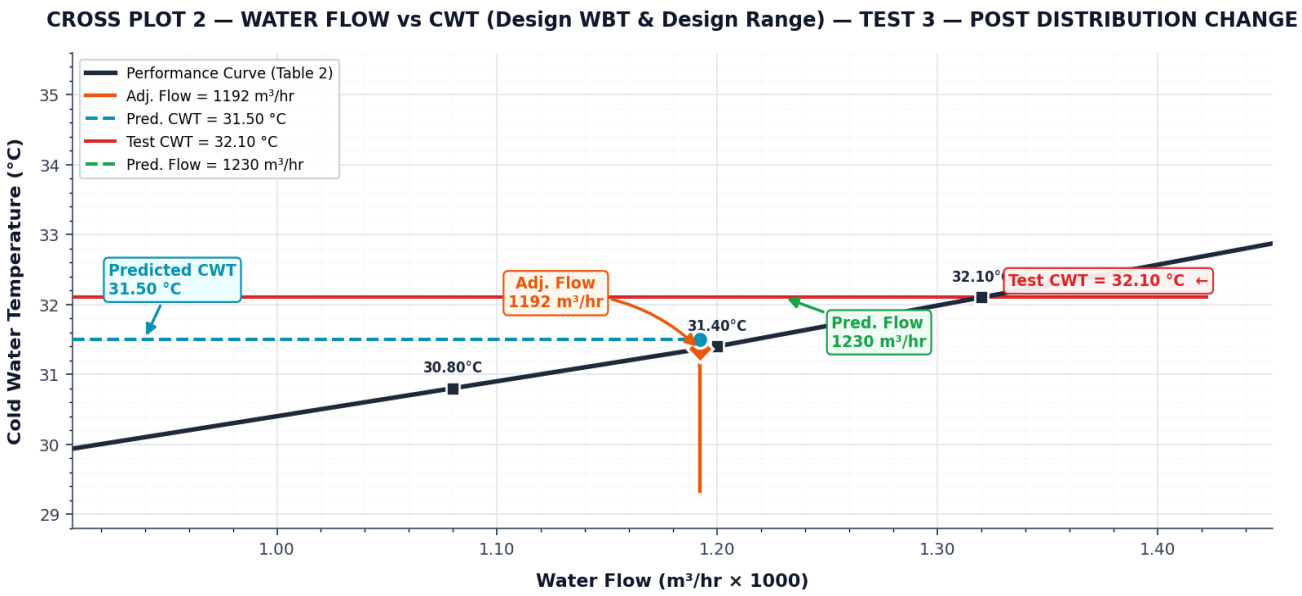

Figure 2: Cross Plot 2 — Water Flow vs CWT (Design WBT & Range) — TEST 3 — POST DISTRIBUTION CHANGE

FINAL ATC-105 RESULTS — TEST 3 — POST DISTRIBUTION CHANGE

STEP 5

FINAL EVALUATION — CWT Shortfall & Capability

Compare the Predicted CWT (from Cross Plot 2 at Q_adj) with the Test CWT. CWT Shortfall = Test CWT – Predicted CWT. A negative shortfall means the tower performed below specification. Capability % = (Q_adj / Predicted Flow) × 100.

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i	Q_adj — Adjusted Water Flow (density corrected)	1192.00 m³/hr
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iii	Test CWT — Recorded during test	31.30 °C
iv	Predicted Water Flow — at Design CWT on Cross Plot 2	1230.00 m³/hr
v	CWT Shortfall = Test CWT – Predicted CWT	-0.20 °C
vi	Cooling Tower Capability = (Q_adj / Pred. Flow) × 100	96.90 %
vii	Air Density Ratio applied	1.00

CONCLUSION — TEST 3 — POST DISTRIBUTION CHANGE

CWT SHORTFALL IN PERFORMANCE	COOLING TOWER CAPABILITY
-0.20 °C	**96.9 %**
Test CWT (31.30°C) vs Predicted CWT (31.50°C)	= 1192 / 1230 × 100

Tower is BELOW specification by 0.20°C — Capability: 96.9%

——— END OF REPORT ———

SS COOLING TOWER CONSULTANTS
www.ssctc.org
15 April 2026